

1

Motivation

SCALE WITH EVIDENCE

- Open-source work often trains standard 7B–70B sizes without revisiting how model, data and optimization should scale.
- Kaplan and Chinchilla reached different model/data prescriptions and left hyperparameter choices underexplored.
- Goal: establish reusable scaling rules and apply them to competitive Chinese–English foundation models.

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Alignment

SFT THEN DPO

- More than one million diverse instruction examples support supervised fine-tuning across language, code, math and reasoning.
- Data ablations expose repetition and knowledge trade-offs; two-stage strategies are explored rather than blindly adding data.
- DPO uses helpfulness and harmlessness preference pairs to improve conversation without materially changing base capabilities.

2

Pre-training

DATA + ARCHITECTURE

- A 2T-token Chinese–English corpus is built through global deduplication, quality filtering and domain remixing.
- 100,015-token BBPE vocabulary; LLaMA-style Pre-Norm, RMSNorm, SwiGLU and RoPE; 67B adds GQA for cheaper inference.
- 7B: 30 layers, 4,096 width; 67B: 95 layers, 8,192 width. Both use 4,096 context and 2T training tokens.

5

Evidence

67B CLOSES THE GAP

- DeepSeek LLM 67B surpasses LLaMA-2 70B on broad benchmarks, especially code, mathematics and reasoning.
- English open evaluation rises from 8.35 after SFT to 8.76 after DPO—above GPT-3.5-turbo's 8.39 and behind GPT-4.
- The chat model performs strongly in both Chinese and English while practical safety tests show largely harmless responses.

3

Scaling laws

COMPUTE-AWARE ALLOCATION

- Optimal batch size increases with compute, while optimal learning rate decreases; power laws predict both.
- Use non-embedding FLOPs/token M instead of parameter count N and model compute precisely as $C = M \times D$.
- Higher-quality data shifts optimal extra compute toward a larger model, so scaling laws are dataset-dependent.

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Limits & meaning

- Knowledge freezes after pre-training; hallucinations and unverified advice remain possible.
- Chinese coverage is incomplete and languages beyond Chinese and English require caution.
- The main contribution is transparent scaling methodology as much as a model release, with code, MoE and stronger data planned next.

DEEPSEEK LLM

DeepSeek LLM

Scaling Open-Source Language Models with Longtermism | 2024

Treat open-model development as a scaling-science problem: jointly study hyperparameters, model/data allocation and data quality, then use the findings to train strong 7B and 67B bilingual models from scratch.

scaling laws

2T tokens

open source